Palatial Pools | Internal Knowledge Hub
Proprietary Technology Technical Specification
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1. Introduction
The Internal Knowledge Hub (IKH) is a purpose-built, centralized digital repository designed to streamline the management, retrieval, and dissemination of operational documentation for Palatial Pool Services Pty Ltd. It enables efficient knowledge sharing, supports regulatory compliance, and enhances workforce training efficacy within the pool service industry.

This document presents an in-depth technical overview of the IKH system, detailing its architecture, data management strategies, security frameworks, and operational protocols. It is intended for engineers, technical managers, prospective buyers, and investors with a focus on understanding the system’s engineering foundation and capabilities.

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2. System Overview

```
### 2.1 Purpose
The IKH serves as an authoritative source for all operational documentation including:
- Training materials and structured learning pathways
- Safety Data Sheets (SDS) with automated parsing for hazard classification
- Standard Operating Procedures (SOPs)
- Equipment manuals and technical guides
- Company policies and compliance documentation

### 2.2 Use Cases
- **Onboarding:** New employees self-navigate structured training modules.
- **Reference:** Experienced technicians access up-to-date procedures and safety protocols.
- **Compliance:** Ensures all documentation aligns with WHS and Australian regulatory standards.

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## 3. Architecture and Data Flow

### 3.1 System Architecture Overview
The IKH employs a multi-tier, modular architecture comprising:
```

+———————–+ +———————–+ +———————–+ | User Interface | <–> | Application Layer | <–> | Data Storage & | | (Web/Mobile) | | (APIs, Business | | Processing Layer | | | | Logic, Search) | | | +———————–+ +———————–+ +———————–+

```
- **User Interface Layer:** Responsive web portal and mobile-optimized client for field access
- **Application Layer:** RESTful API services, business logic including document management, search, and metadata extraction
- **Data Storage Layer:** Encrypted relational database and object storage for documents and metadata

### 3.2 Data Flow Diagram

```mermaid
flowchart TD
    A[User Request] --> B[Authentication & Authorization]
    B --> C[Document Query/Upload API]
    C --> D{Request Type?}
    D -->|Search| E[Full-Text Search Engine]
    D -->|Upload| F[Document Parser & Metadata Extractor]
    F --> G[Version Control Module]
    E --> H[Results]
    G --> I[Database Storage]
    I --> H
    H --> J[User Interface: Display Results]
```

3.3 Data Flow Description

- User initiates a request via UI; credentials validated through Role-Based Access Control (RBAC) service.
- Queries routed to search engine or uploads handled by parser modules.
- Document uploads trigger metadata extraction, including SDS parsing with hazard classification.
- Version control ensures all document changes are tracked and the latest versions served.
- Data stored in encrypted databases with audit logs for compliance and traceability.

```

4. Core Functional Components

4.1 Document Repository

- Stores all operational documents in PDF and structured text formats.
- Supports hierarchical categorization for intuitive navigation.
- Employs metadata tagging for enhanced filtering capabilities.

4.2 Search Engine

- Implements an inverted index full-text search using Apache Lucene or equivalent.
- Supports Boolean, phrase, and fuzzy search queries.
- Tag-based and category filters enable precise document retrieval.

4.3 SDS Hazard Parser

- Utilizes Natural Language Processing (NLP) techniques to parse SDS documents.
- Extracts key information fields: hazard classifications (GHS codes), first aid measures, storage conditions.
- Outputs structured JSON metadata conforming to Globally Harmonized System (GHS) standards.

4.4 Version Control System

- Git-like versioning model tailored for document revision tracking.
- Stores incremental diffs and supports rollback to previous versions.
- Maintains audit trail with timestamps, editor identities, and change summaries.

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5. Document Management and Version Control

5.1 Document Lifecycle

- **Creation:** Documents uploaded via UI or API with mandatory metadata fields.

- **Review:** Assigned reviewers validate documents before publication.
- **Publication:** Approved documents become accessible based on user roles.
- **Revision:** Updates trigger version increments; superseded versions archived but accessible for audit.

5.2 Version Control Implementation Details

- Backend employs a content-addressable storage model referencing SHA-256 hashes.
- Metadata stored in PostgreSQL with JSONB columns for flexible schema evolution.
- Version branches allow parallel document editing workflows for different teams.

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6. Search Engine and Metadata Extraction

6.1 Search Engine Architecture

- Indexing pipeline processes new and updated documents asynchronously.
- Index shards distributed across nodes for scalability and fault tolerance.
- Query engine supports weighted ranking based on recency, document type, and user access patterns.

6.2 SDS Metadata Extraction Algorithm

Parsing Steps:

1. **Document Preprocessing:**
    - OCR applied to scanned PDFs using Tesseract with language model tuned for chemical terminology.
    - Text normalized (case folding, punctuation removal).
2. **Section Segmentation:**

    - Regex-based heuristics detect standard SDS section headings (e.g., “Hazard Identification”, “First Aid Measures”).
    - Sections extracted as discrete text blocks.
3. **Entity Recognition:**
    - Named Entity Recognition (NER) models identify hazard classes, chemical names, storage recommendations.
    - Mapping to standard GHS codes using lookup tables.
4. **Structured Output:**
    - Extracted data serialized into JSON conforming to ISO 11014 SDS data format guidelines.

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7. Security and Compliance

7.1 Role-Based Access Control (RBAC)

- User roles defined with least privilege principles:
    - **Technician:** Read-only access to operational documents.
    - **Supervisor:** Read/write access to SOPs and training materials.
    - **Administrator:** Full system privileges including user management.

7.2 Data Encryption and Integrity

- Data at rest encrypted using AES-256 symmetric encryption via PostgreSQL Transparent Data Encryption (TDE).
- Data in transit secured by TLS 1.3 protocol with mutual authentication for API endpoints.
- Document integrity verified through SHA-256 hashing; hashes stored and verified on each access.

7.3 Audit Trails and Logging

- Immutable audit logs maintained in append-only format using write-once storage.
- Logs include user identity, timestamps, action types, and document versions accessed or modified.
- Supports compliance with Australian Work Health and Safety (WHS) regulations and data retention policies.

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8. Infrastructure and Deployment

8.1 Hosting Environment

- Hosted on Australian-based cloud infrastructure with data residency guarantees (e.g., AWS Sydney Region with data sovereignty compliance).
- Infrastructure as Code (IaC) using Terraform for consistent environment provisioning.

8.2 System Scalability

- Microservices deployed in Docker containers orchestrated by Kubernetes (K8s) clusters.
- Horizontal scaling of API and search services based on load.

8.3 Mobile Optimization

- Responsive UI built with React.js and Progressive Web App (PWA) standards.
- Offline caching of recent documents using IndexedDB for field use without network connectivity.

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9. Integration and API Specifications

9.1 REST API Overview

- API base URL: `https://api.palatialpools.com/knowledgehub/v1`
- Authentication: OAuth 2.0 with JWT access tokens.

9.2 Key Endpoints

Endpoint	Method	Description	Request/Response Format
`/documents`	GET	Retrieve documents with filtering options	Query params; JSON response
`/documents/{id}`	GET	Retrieve single document metadata/content	JSON metadata, Base64 PDF content
`/documents`	POST	Upload new document	Multipart/form-data, JSON metadata
`/documents/{id}/versions`	GET	List document version history	JSON array
`/search`	POST	Full-text search with filters	JSON query; JSON results

9.3 Data Formats

- **Document Metadata JSON Schema (excerpt):**

```
{
  "documentId": "string",
  "title": "string",
  "category": "string",
  "tags": ["string"],
  "version": "string",
  "createdBy": "string",
  "createdAt": "ISO8601 datetime",
  "hazardInfo": {
    "ghsClassification": ["string"],
    "firstAidMeasures": ["string"],
    "storageRequirements": ["string"]
  }
}
```

- **Search Query Example:**

```
{
  "query": "chlorine hazard",
  "filters": {
    "category": ["SDS", "Safety"],
    "tags": ["chemical"]
  },
  "pagination": {
    "page": 1,
    "perPage": 20
  }
}
```

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10. Standards and Regulatory Compliance

Aspect	Standard / Regulation	Description
Safety Data Sheets (SDS)	Globally Harmonized System (GHS) ISO 11014	Standardized SDS format and hazard classification
Data Encryption	NIST SP 800-57, AES-256	Cryptographic standards for data protection
Network Security	TLS 1.3 (RFC 8446), OAuth 2.0 (RFC 6749)	Secure data transmission and authentication
Workplace Health & Safety	Australian WHS Act 2011	Compliance with workplace safety documentation
Accessibility	WCAG 2.1 AA	Web content accessibility guidelines
Data Sovereignty	Australian Privacy Act 1988	Ensures data residency and privacy compliance

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11. Glossary of Terms

- **AES-256:** Advanced Encryption Standard with 256-bit keys for data encryption.
- **GHS:** Globally Harmonized System of Classification and Labelling of Chemicals.
- **NLP:** Natural Language Processing, a subfield of AI focused on understanding human language.
- **OAuth 2.0:** Authorization framework enabling applications to obtain limited access tokens.
- **PWA:** Progressive Web App, a web application that behaves like a native app.
- **RBAC:** Role-Based Access Control, restricting system access based on user roles.
- **SDS:** Safety Data Sheet, a document detailing chemical safety and handling information.
- **SHA-256:** Secure Hash Algorithm producing a 256-bit hash used for data integrity.

- **TLS 1.3:** Transport Layer Security protocol version 1.3, providing secure communications.

Appendix A: Sample JSON SDS Metadata Output

```
{
  "documentId": "SDS-2024-0001",
  "title": "Chlorine Gas Safety Data Sheet",
  "category": "SDS",
  "tags": ["chemical", "hazard", "chlorine"],
  "version": "1.2",
  "createdBy": "safety_team",
  "createdAt": "2024-05-10T09:30:00Z",
  "hazardInfo": {
    "ghsClassification": [
      "Flammable gas Category 1",
      "Acute toxicity Category 3",
      "Skin corrosion Category 1B"
    ],
    "firstAidMeasures": "In case of inhalation, move to fresh air
immediately. Seek medical attention if symptoms develop.",
    "storageRequirements": "Store in a cool, well-ventilated area away from
incompatible materials."
  }
}
```

Appendix B: Security Protocol Stack

```
+----------------------------+
| Application Layer          |
| - OAuth 2.0 Authentication |
| - Role-Based Access Control|
+----------------------------+
             |
+----------------------------+
| Transport Layer            |
| - TLS 1.3 Encryption       |
+----------------------------+
             |
+----------------------------+
| Network Layer              |
| - Firewall & IDS/IPS       |
+----------------------------+
             |
+----------------------------+
| Data Layer                 |
| - AES-256 Data Encryption  |
| - SHA-256 Hashing          |
+----------------------------+
```

Contact and Support

For further technical details, deployment assistance, or integration support, please contact:

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